
Seeing Isn't Structuring: Multimodal Failure Modes in Vision-Language Models

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Abstract

Strong performance on visual questions does not necessarily mean that a vision-language model has recovered the structure needed to support its answer. We introduce SL-7, a controlled Snakes & Ladders diagnostic that separates local visual perception from structured multimodal reasoning. Seven open-weight VLMs are evaluated on seven failure modes spanning spatial direction, hidden-layout reconstruction, endpoint grounding, perception-to-action transfer, multi-image comparison, conditioned counting, and multi-source composition, with prerequisite calibration for each family. Hidden-layout reconstruction is the most consistent failure: five number-calibrated models answer all 50 visible controls correctly but only 1 of 50 corresponding hidden-row probes. Direction reversal is more model dependent, although 20 of 21 errors follow the same $+1$ shortcut. Qwen3-VL-4B further exhibits endpoint hallucination, filtered-counting, and composition failures. These results show that accurate local perception can coexist with brittle use of latent visual structure, and that prerequisite-aware evaluation is essential for distinguishing higher-order reasoning failures from basic perceptual limitations.

1 Introduction

Modern VLMs combine OCR, recognition, document understanding, and multimodal reasoning in compact open-weight checkpoints (Bai et al., 2025b,a; Abdin et al., 2024; Wang et al., 2025; Marafioti et al., 2025; Gemma Team, 2025). Broad benchmarks such as MMBench and MMMU measure useful task-level coverage (Liu et al., 2024; Yue et al., 2024), but aggregate accuracy often mixes perception, localization, relational reasoning, counting, comparison, and arithmetic. A correct answer therefore does not reveal which intermediate capability was actually used.

We study a narrower question: *if a model can perceive the ingredients of a visual problem, can it still use them when the task requires latent structure?* Snakes & Ladders provides a controlled setting for this question. Its boustrophedon numbering separates physical direction from numerical direction, while snakes, ladders, dice, and paired boards support endpoint grounding, state transitions, counting, comparison, and composition.

We introduce SL-7, a 170-query diagnostic with seven families and matched controls. Seven open-weight VLMs are evaluated with fixed prompts, greedy decoding, strict exact-match scoring, and prerequisite-aware calibration. Three findings dominate. First, hidden-layout reconstruction collapses across every number-calibrated model: 50/50 visible controls versus 1/50 hidden probes. Second, direction-linearization is model dependent, although 20/21 B1 errors share the same $T + 1$ shortcut. Third, Qwen3-VL-4B, the only model eligible for all seven families, also confirms failures in endpoint absence, conditioned counting, and multi-source composition.

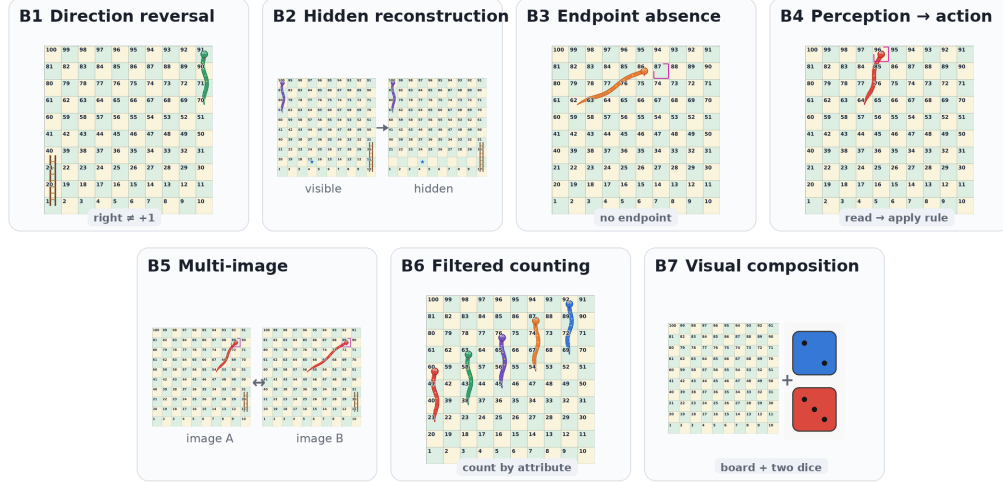


Figure 1: Overview of the seven SL-7 failure-mode families. Each panel shows a representative stimulus from the same controlled visual world: direction reversal (B1), hidden reconstruction (B2), endpoint absence (B3), perception-to-action transfer (B4), multi-image comparison (B5), conditioned counting (B6), and multi-source composition (B7).

Contributions. We provide (i) a single controlled visual environment spanning seven multimodal interfaces, (ii) family-specific prerequisite gating that separates perception failure from reasoning failure, and (iii) cross-model error-signature analysis rather than a single aggregate score.

2 Related work

Controlled benchmarks such as CLEVR, VSR, and Winoground use synthetic or contrastive examples to isolate compositional and relational failures (Johnson et al., 2017; Liu et al., 2023; Thrush et al., 2022). Hallucination benchmarks such as POPE and HallusionBench similarly show that visual access does not prevent unsupported or inconsistent predictions (Li et al., 2023; Guan et al., 2023). More generally, shortcut learning describes reliance on predictive but non-robust cues rather than intended structure (Geirhos et al., 2020). SL-7 follows this diagnostic tradition but keeps all probes in one visual world, uses matched controls, and records predicted error signatures.

3 The SL-7 benchmark

3.1 Design

The benchmark uses a standard 10×10 board with boustrophedon numbering: odd rows increase left-to-right and even rows increase right-to-left. The final set contains 150 scored items and 20 calibration items. Six families contain 20 scored items each; B7 contains 30 because every scenario includes two component reads and one composition query. Figure 1 shows representative stimuli.

3.2 Confirmation and calibration

Low probe accuracy alone is insufficient. The fixed confirmation rules require strong control performance plus a probe drop and, for B1–B6, a predicted error signature. In compact form: B1 requires control $\geq 8/10$, probe $\leq 5/10$, gap $\geq 30pp$, and $\geq 60\% T + 1$ errors; B2 requires visible $\geq 9/10$, hidden $\leq 5/10$, gap $\geq 40pp$, and $\geq 60\%$ structural-shortcut errors; B3 requires positive endpoints $\geq 8/10$, absence $\leq 5/10$, and $\geq 60\%$ positive hallucinations; B4 requires destination reads $\geq 8/10$ and $\leq 50\%$ action accuracy on prerequisite-correct probes; B5 requires obvious controls $\geq 3/4$, subtle accuracy $\leq 8/16$, and position lock in at least $5/8$ pairs; B6 requires total-count

Table 1: Seven benchmark families. The final column gives the targeted systematic failure, not merely a wrong answer.

Family	Control / probe	Targeted signature or criterion
B1 Direction reversal	Right-neighbor relation on matched rows	On even rows, answer $T + 1$ instead of correct $T - 1$
B2 Hidden reconstruction	Visible target vs. same target with row numbers hidden	Error matches a predefined structural alternative
B3 Endpoint absence	True endpoints vs. nearby/body-crossing empty cells	Hallucinate a positive endpoint on an absence probe
B4 Perception→action	Read destination, then apply the game rule	Stay on the landing square despite correct destination read
B5 Multi-image position	Obvious vs. subtle order-swapped image pairs	Same A/B response after image order is reversed
B6 Conditioned counting	Count all objects vs. attribute-filtered count	Return the unfiltered total
B7 Multi-source composition	Read two dice separately, then compose with board state	Large component-to-composition accuracy gap

Table 2: Model inventory and calibration. Bold values pass the corresponding gate; “Elig.” is the number of headline-eligible families.

Model	Params	Number	Endpoint	Die	Multi-img.	Elig.
Qwen3-VL-4B	4.44B	8/8	3/4	4/4	3/4	7
Qwen3-VL-2B	2.13B	8/8	4/4	4/4	2/4	5
Qwen2.5-VL-3B	3.75B	8/8	0/4	3/4	2/4	2
Phi-3.5-Vision	4.15B	8/8	0/4	3/4	2/4	2
InternVL3.5-4B	4.73B	7/8	2/4	4/4	0/4	2
SmolVLM2-2.2B	2.25B	2/8	2/4	3/4	2/4	0
Gemma-3-4B	4.30B	0/8	1/4	4/4	2/4	0

controls $\geq 8/10$, filtered accuracy $\leq 5/10$, and $\geq 60\%$ unfiltered-total errors; B7 requires both dice correct in at least 8/10 scenarios, composition $\leq 50\%$, and a component-to-composition gap $\geq 40 pp$.

Calibration tests basic legibility with 8 number-reading, 4 endpoint, 4 die, and 4 obvious multi-image items. Pass marks are 7/8, 3/4, 4/4, and 3/4. Eligibility is family specific: B1–B2 require number calibration; B3 endpoint; B4/B6 number+endpoint; B5 number+endpoint+multi-image; B7 number+die+multi-image. Thus a model is not credited with a higher-order failure mode when its prerequisite visual skill is absent.

3.3 Generation and scoring

Boards, endpoints, dice, and paired-image changes are synthetically generated with fixed rendering and automated geometry/answer-key checks. Every query uses a fresh conversation. Primary scoring is strict exact match: an integer response must contain exactly one valid integer, and a label response exactly one allowed label after normalization. We retain raw outputs; lenient parsing is diagnostic only. Exact McNemar tests are used for matched control/probe outcomes and Wilson intervals for accuracy.

4 Models and inference

We evaluate seven compact open-weight VLMs. All runs use greedy decoding, batch size 1, fresh context per item, and `max_new_tokens=64`. Most models run in FP16 on NVIDIA T4 hardware; InternVL3.5-4B uses GPU/CPU dispatch, Phi-3.5-Vision uses its native crop settings, and Gemma-3-4B uses FP32 for stable generation. These execution choices do not change benchmark inputs or scoring.

Table 3: Cross-model headline status. B1/B2 have five eligible models; the other families have fewer and should not be read as equally broad cross-model claims.

Model	B1	B2	B3	B4	B5	B6	B7
Qwen3-VL-4B	–	△	✓	–	△	✓	✓
Qwen3-VL-2B	✓	✓	–	–	NE	–	NE
Qwen2.5-VL-3B	✓	△	NE	NE	NE	NE	NE
Phi-3.5-Vision	–	✓	NE	NE	NE	NE	NE
InternVL3.5-4B	–	✓	NE	NE	NE	NE	NE
SmolVLM2-2.2B	NE	NE	NE	NE	NE	NE	NE
Gemma-3-4B	NE	NE	NE	NE	NE	NE	NE
Eligible models	5	5	2	2	1	2	1
Confirmed models	2	3	1	0	0	1	1

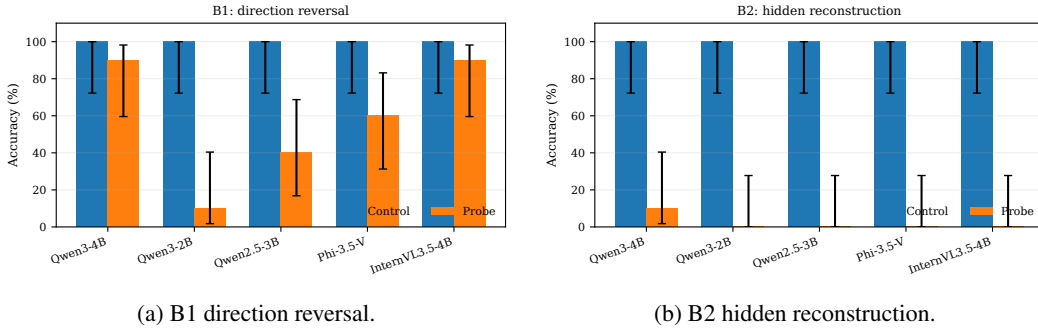


Figure 2: Control and probe accuracy for the five B1/B2-eligible models. Error bars are 95% Wilson intervals.

5 Results

5.1 Cross-model overview

Table 3 summarizes the final interpretation. ✓ denotes confirmed, △ a strong secondary effect, “–” eligible but not confirmed, and NE prerequisite-ineligible.

5.2 B2: a robust hidden-structure failure

B2 is the strongest result. All five number-calibrated models answer every visible target correctly (**50/50**), but hidden-row reconstruction succeeds only **1/50** times. Three models satisfy the full shortcut rule: Qwen3-VL-2B (7/10 error matches), Phi-3.5-Vision (6/10), and InternVL3.5-4B (9/10). Qwen3-VL-4B still falls from 10/10 to 1/10 but has only 3/9 signature matches; Qwen2.5-VL-3B falls from 10/10 to 0/10 with 4/10 matches. Thus the behavioral collapse is universal among eligible models, while the exact shortcut mechanism varies.

5.3 B1: a shared error type with model-dependent frequency

The same five models are perfect on matched B1 controls (50/50) but score 29/50 on direction-reversal probes. Qwen3-VL-2B (1/10) and Qwen2.5-VL-3B (4/10) satisfy the full B1 rule; Phi-3.5-Vision scores 6/10, while Qwen3-VL-4B and InternVL3.5-4B each score 9/10. Yet **20/21 probe errors (95.2%)** equal the predicted $T + 1$ shortcut. The shortcut therefore appears broadly shared as an error mode, even though its frequency differs sharply by model.

Table 4: Detailed B1/B2 results. “Sig.” is signature matches among probe errors; p is the exact McNemar test.

Model	Fam.	Control	Probe	Gap (pp)	Sig.	p
Qwen3-VL-4B	B1	10/10	9/10	10	1/1	1.000
	B2	10/10	1/10	90	3/9	0.0039
Qwen3-VL-2B	B1	10/10	1/10	90	9/9	0.0039
	B2	10/10	0/10	100	7/10	0.0020
Qwen2.5-VL-3B	B1	10/10	4/10	60	6/6	0.0313
	B2	10/10	0/10	100	4/10	0.0020
Phi-3.5-Vision	B1	10/10	6/10	40	3/4	0.125
	B2	10/10	0/10	100	6/10	0.0020
InternVL3.5-4B	B1	10/10	9/10	10	1/1	1.000
	B2	10/10	0/10	100	9/10	0.0020

Table 5: Complete seven-family profile for Qwen3-VL-4B.

Family	Control/comp.	Probe	Gap (pp)	Signature	Status
B1 Direction	10/10	9/10	10	1/1	Not confirmed
B2 Hidden	10/10	1/10	90	3/9	Secondary
B3 Endpoint absence	10/10	0/10	100	10/10	Confirmed
B4 Perception→action	5/10	1/5	30	1/4	Not confirmed
B5 Multi-image	3/4	9/16	18.8	lock 7/8	Secondary
B6 Filtered count	8/10	0/10	80	6/10	Confirmed
B7 Composition	20/20	1/10	90	–	Confirmed

5.4 Remaining families

Qwen3-VL-4B is the only model that passes all four calibration gates, so it provides the cleanest complete seven-family profile (Table 5). B3 is confirmed: all 10 positive endpoints are recognized, but all 10 absence probes hallucinate a positive endpoint. B6 is also confirmed: unfiltered counting is 8/10 while filtered counting is 0/10, with 6/10 errors repeating the unfiltered total. B7 gives a 20/20 component-read score but only 1/10 strict compositions, confirming a component-to-composition gap.

B4 is a negative result because destination perception itself is below the prerequisite threshold (5/10 for Qwen3-VL-4B; 3/10 for Qwen3-VL-2B), so an action-transport failure cannot be isolated. B5 is secondary rather than confirmed: Qwen3-VL-4B passes obvious multi-image controls (3/4), shows 7/8 position-locked subtle pairs and 15/16 A responses, but scores 9/16—one item above the fixed failure threshold. Several other models show extreme A/B collapse, but they fail obvious multi-image calibration and are therefore not evidence for the higher-order B5 mechanism.

Why calibration matters. SmolVLM2 and Gemma produce dramatic raw gaps on some downstream tasks but fail the prerequisite number/endpoint gates, so none of their seven families are headline eligible. Likewise, Qwen3-VL-2B can contribute to B1/B2/B3/B4/B6 but not B5/B7 because it fails obvious multi-image calibration. This distinction prevents basic perceptual failure from being mislabeled as structured reasoning failure.

6 Discussion and limitations

The central result is not simply that models fail on a board game. The five B2-eligible models can read visible marked cells perfectly, yet nearly all fail when the answer must be reconstructed from the board’s latent organization. This separates local visual recognition from use of the relation that generated the visible symbols. B1 adds a complementary finding: error *type* can be stable even when error *frequency* is not. Across models, almost every B1 mistake is the same $T + 1$ shortcut, while probe accuracy ranges from 1/10 to 9/10.

The Qwen3 pair also shows that scale does not remove failures uniformly. Moving from 2B to 4B greatly improves B1 (1/10 to 9/10) but leaves B2 near floor (0/10 to 1/10); at the same time, the stronger 4B model becomes eligible for additional tests and exposes confirmed B3, B6, and B7 failures.

The study is deliberately controlled, which is both a strength and a limitation. The domain is synthetic; most families contain only 10 matched scenarios; strict parsing can penalize verbose answers; and family coverage is unequal because only one model passes every calibration gate. The strongest claims are therefore B1/B2, where five models are eligible. The observed signatures are behavioral evidence for shortcuts, not mechanistic evidence about internal representations. Extending the same interfaces to documents, maps, diagrams, and natural images is an important next step.

7 Conclusion

Across seven compact VLMs, SL-7 reveals a consistent gap between seeing local visual evidence and using latent structure. The most robust effect is hidden reconstruction: five calibrated models move from 50/50 visible controls to 1/50 hidden probes. Direction reversal is less universal, but 20/21 errors share the same numerical-shortcut signature. Qwen3-VL-4B additionally confirms endpoint-absence hallucination, conditioned counting failure, and multi-source composition failure. These results support prerequisite-aware multimodal evaluation that asks not only whether a model can perceive isolated facts, but whether those facts remain usable when they must be reconstructed, filtered, compared, or composed.

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